

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- 파생분포 $Y=g(X)$: CDF 먼저 $\rightarrow$ 미분, 선형 $Y=aX+b \rightarrow f_Y=(1/|a|)f_X((y-b)/a)$
- $Z=X+Y$ (독립) 컨볼루션, 정규+정규=정규
- $\text{Cov}(X,Y)=E[XY]-E[X]E[Y]$, $\text{Var}(X+Y)=\text{Var}X+\text{Var}Y+2\text{Cov}$

문항 목차 (총 11문항)

- [1] L07 · IPSRP 4.4.4 — Further1 변환·공분산
- [2] L07 · IPSRP 4.4.5 — Further1 변환·공분산
- [3] L07 · IPSRP 4.4.6 — Further1 변환·공분산
- [4] L07 · IPSRP 5.4.27 — Further1 변환·공분산
- [5] L07 · IPSRP 5.4.28 — Further1 변환·공분산
- [6] L07 · PSP 4.6.1 — Further1 변환·공분산
- [7] L07 · PSP 4.6.2 — Further1 변환·공분산
- [8] L07 · PSP 4.6.5 — Further1 변환·공분산
- [9] L07 · PSP 5.7.3 — Further1 변환·공분산
- [10] L07 · PSP 5.8.1 — Further1 변환·공분산
- [11] 복습 L03 · PSP 3.2.1 — 이산RV1

Problem 4

Let X be a *uniform*(0, 1) random variable, and let $Y = e^{-X}$.

- a. Find the CDF of Y .
 - b. Find the PDF of Y .
 - c. Find EY .
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Problem 5

Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} \frac{5}{32}x^4 & 0 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

and let $Y = X^2$.

- Find the CDF of Y .
- Find the PDF of Y .
- Find EY .

Problem 6

Let $X \sim \text{Exponential}(\lambda)$, and $Y = aX$, where a is a positive real number. Show that

$$Y \sim \text{Exponential}\left(\frac{\lambda}{a}\right).$$

Problem 27

Let X and Y be two independent $Uniform(0, 1)$ random variables, and $Z = \frac{X}{Y}$. Find the CDF and PDF of Z .

— 풀이 —

Problem 28

Let X and Y be two independent $N(0, 1)$ random variables, and $U = X + Y$.

- Find the conditional PDF of U given $X = x$, $f_{U|X}(u|x)$.
- Find the PDF of U , $f_U(u)$.
- Find the conditional PDF of X given $U = u$, $f_{X|U}(x|u)$.
- Find $E[X|U = u]$, and $Var(X|U = u)$.

4.6.1 ● The peak temperature T , as measured in degrees Fahrenheit, on a July day in New Jersey is the Gaussian $(85, 10)$ random variable. What is $P[T > 100]$, $P[T < 60]$, and $P[70 \leq T \leq 100]$?

4.6.2 ● What is the PDF of Z , the standard normal random variable?

— 풀이 —

4.6.5● Your internal body temperature T in degrees Fahrenheit is a Gaussian ($\mu = 98.6, \sigma = 0.4$) random variable. In terms of the $\Phi(\cdot)$ function, find $P[T > 100]$. Does this model seem reasonable?

5.7.3● A random ECE sophomore has height X (rounded to the nearest foot) and GPA Y (rounded to the nearest integer). These random variables have joint PMF

$P_{X,Y}(x,y) \mid$	$y = 1$	$y = 2$	$y = 3$	$y = 4$
$x = 5$	0.05	0.1	0.2	0.05
$x = 6$	0.1	0.1	0.3	0.1

Find $E[X + Y]$ and $\text{Var}[X + Y]$.

5.8.1● X and Z are independent random variables with $E[X] = E[Z] = 0$ and variance $\text{Var}[X] = 1$ and $\text{Var}[Z] = 16$. Let $Y = X + Z$. Find the correlation coefficient ρ of X and Y . Are X and Y independent?

3.2.1● The random variable N has PMF

$$P_N(n) = \begin{cases} c(1/2)^n & n = 0, 1, 2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[N \leq 1]$?